



Sound Pitch Synthesizer

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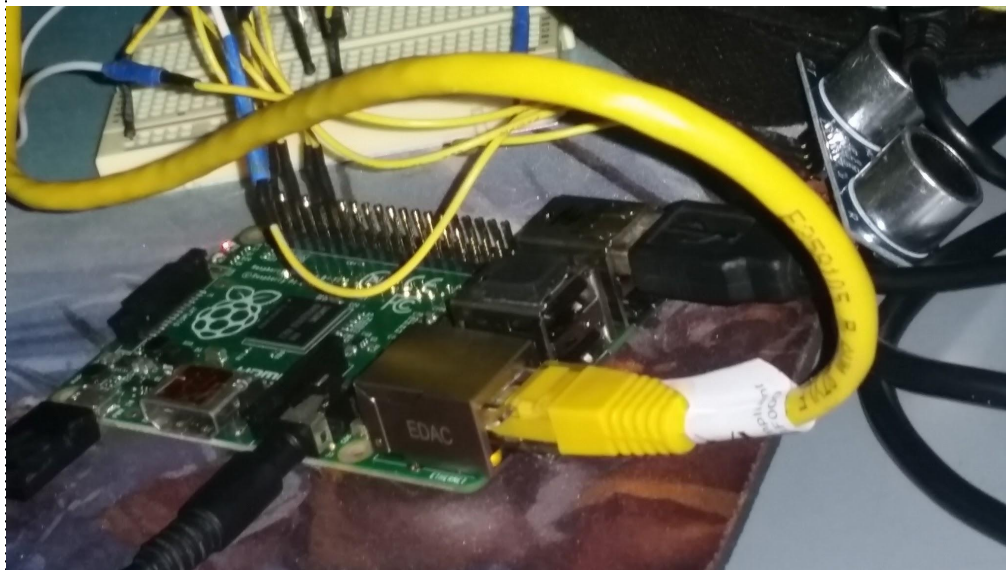
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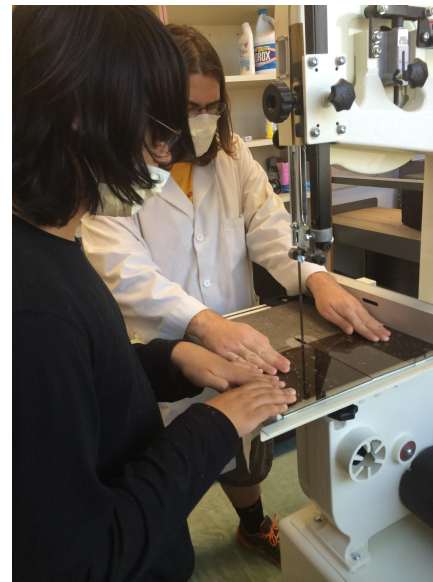
Abstract: During the sound synthesizer experience the students each participated in an aspect of the final product, either programming or physical components. For the physical component aspect, the materials were gathered and then attempts at creating a housing for the synthesizer were made to a success. On the programming end, the code was done by using libraries that were made and were then implemented for our purposes.

Introduction: We began by figuring out how the raspberry pi fit together with the others components. Then Mario began the programming aspect and Christian began the physical aspect. Physically the objective was a cube. Programming the objective was to make a code, which, from the detection of a certain height would output a sound.

Programming Method: The code within the Pi consists of two portions. One is the sound framework. Mario used a library called PyAudio, which enabled calling functions that would output any frequency given in its parameters. The second portion consisted of the code necessary to detect the distance coming from the ultrasonic sensor. This was done using mathematics and a little knowledge on how the ultrasonic sensor worked. First, it sends a sound wave, that, when it hits an object, it bounces back for detection. The program found the time it took to bounce back, and based on that, we are able to find the distance. $\text{Time} * \text{Speed of sound} = \text{Distance traveled} * 2$. Unification of these two portions was what made it possible to create a sound frequency based on the height detected by the ultrasonic sensor.



Physical Method: Acquiring materials and being able to use the machine shop in order to cut and drill them. The main material used ended up being plexiglass. It was difficult to manage and cut into pieces desired. The holes needed for slots constantly gave us issues by chipping or fracturing.

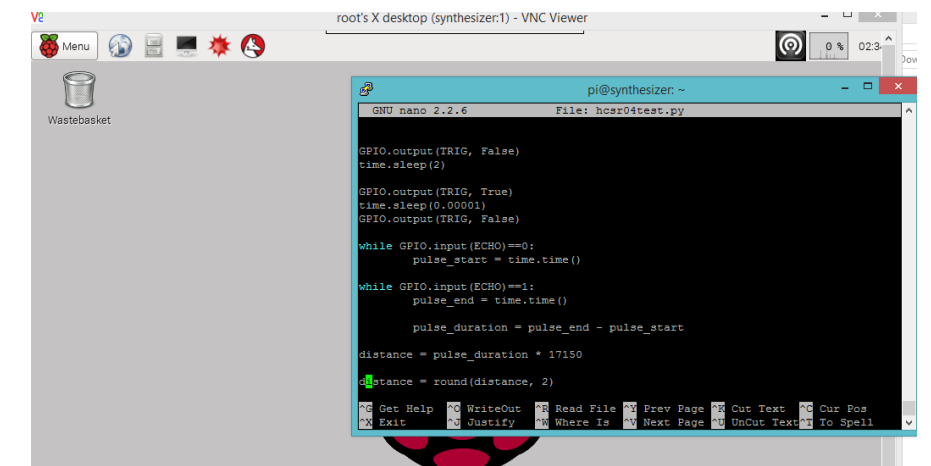


Construction process: Initially we needed to gather materials to build a cube. The material used is Bronze Acrylic Plexiglass. Being a rather fragile material we, had different trials and errors throughout the process of construction. The first part was creating six panels to fit together to make a 7x7 cube. so getting dimensions for each plate and sizing them was the first challenge. Following up We needed to be able to solve issue of how cords were to enter as well as pi placement. The decision was to make holes to allow for the passing of the cords. The main issue here was the breaking of the material due to high strain. So trial and error was done to achieve desired results.

Results: A housing was created as well as a object capable of producing sound from height detection.

Conclusion: The Sound Pitch Synthesizer is a work in progress as we try to smooth out any outlying kinks in code to make the sound flow. There are other application possibilities as well with this style of technology.

Possible other Applications: A disk jockey could implement this system for their own means to make different songs or beats play by waving his hands at a height. this could be also used as something for a security code by making a certain height pattern and reproducing it. Though any of these idea would involve brand new coding.



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Sound Pitch Synthesizer

Mario Canul and Christian Innela, Mentor: Dr. Buddhi Rai

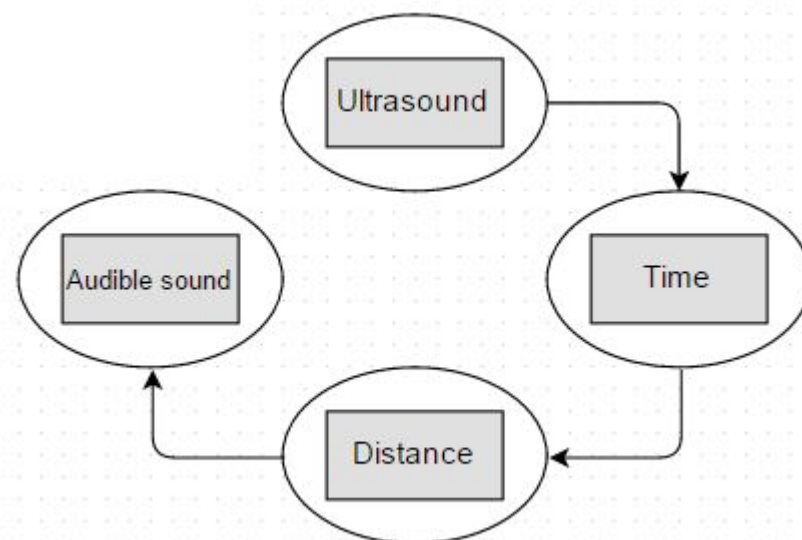
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Abstract: Our project was to create an audible sound out of a distance. So we built and programmed a device that converts the time it takes for an ultrasound to reach a target and bounce back, readily converting it to a distance, then to an audible sound.

Function Map:

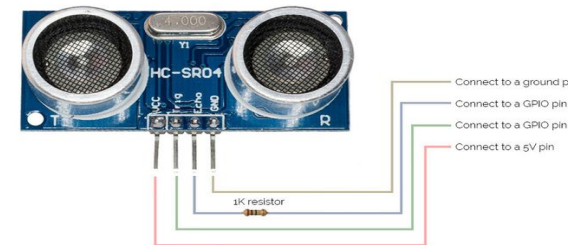


This was accomplished by designing a simple electronic circuit, writing a code to interpret the time, and creating a function that produces an audible sound using the distance calculated.

Device Assembly:

Raspberry Pi B+ w/ 8 GB SD card, HCSR04 Ultrasonic Sensor, Breadboard, Copper Wires, 1k Ohm resistor, Speakers, 2.1 Amp. Micro USB Power Cable, Plexiglass, Laptop configured to local LAN

Circuitry: The sensor component is what makes this device have hands-free functionality. In order for the whole electronic system to work, we needed to create a simple yet useful electrical circuit that will provide the means to send the information of the HCSR04 sensor to the Raspberry Pi for signal interpretation. The circuit is as follows:



Programming Scheme: First, in order to convert the time into a distance in the programming, we needed to use a mathematical relation that uses time and the speed of sound, which is a constant at 343 m/s. The distance is defined as:

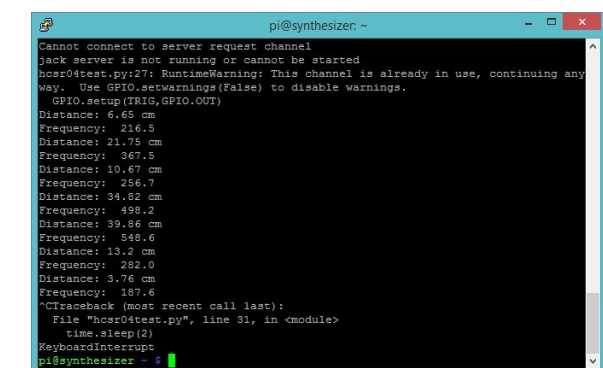
$$d = \frac{(t_f - t_i) v_{sound}}{2}$$

where “ d ” is distance, “ t ” is time, and “ v ” is the speed of sound.

Second, we needed the Raspberry Pi to actually create a sound. This was done through a library called PyAudio that creating an audio stream that we can write information to, such as the information required to make a sine wave. The structure of the sound function creates a frequency to be written to the audio stream.

$$\lambda = 10d + x_{displacement}$$

Results: The sound pitch synthesizer works as designed. The frequencies produced were accurate as we tested them using a sound sensor available in the Physics lab at UHMC. The height was also accurate from testing a laser device. A housing was mounted to protect the circuitry and hold the sensor in place.



The longer the distance traveled by the ultra-sound, the higher the frequency of the audible sound. This could be used as an “audio” measurement of a distance.

Conclusion: There are other application possibilities with this hands-free technology. A disk jockey could implement this system for his/her own means to make different songs or beats play by waving their hands at a height. This could be also used as technology for a security system by making a certain length pattern and reproducing it, or as an alternative sight method for the vision impaired. New musical instruments could be created from this concept. There may be applications in rescue situations to tell how far an object is through a sound when visibility is not available.

